

Smart contract layer: Any request for transaction handling is validated by executing the business logic in the Chaincode function and is taken care of by this layer.

Communications/protocol layer: This layer handles the peer-to-peer message-passing between each of the participating nodes in the platform, in order to manage the transactions in the shared ledger copy in the blocks.

Data layer: All the data handling activities including the enrichment of transactions, audit trails of message processing and transaction states are managed in this layer. It also has crypto abstraction, which enables data to be securely stored by using different crypto algorithms. This layer ensures the identity services layer, establishing the root of trust in the blockchain instance.

Identity services layer: Membership registration services are handled in this layer to ensure authorisation and authentication of member access, across nodes in the platform.

API layer: All service interface capabilities to third party tools and frameworks are enabled by this layer to handle synchronous calls of requests and responses to the blockchain platform.

Policy services layer: This enables the governing principles, and caters to the business policy of the underlying blockchain platform.

The open source Hyperledger software development community

Hyperledger project development and consensus management is governed by the Hyperledger Architecture Working Group. This serves as a cross-project forum to bring together the community members and architects to develop the community and implement the underlying framework. For example, consensus can be implemented with different lottery based algorithms

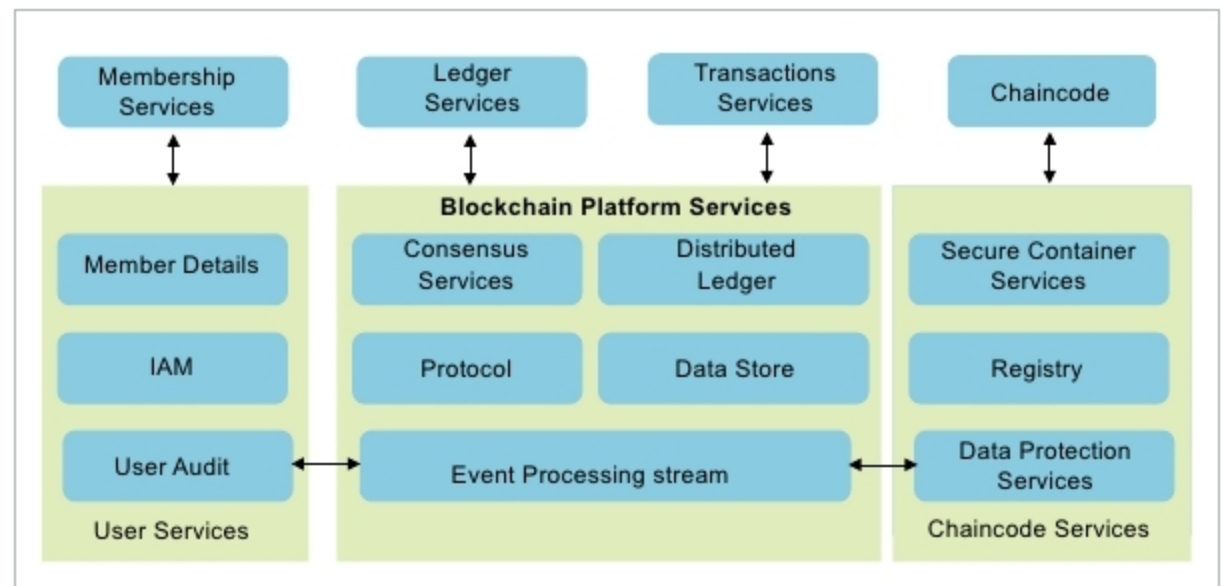


Figure 1: Services in Hyperledger architecture

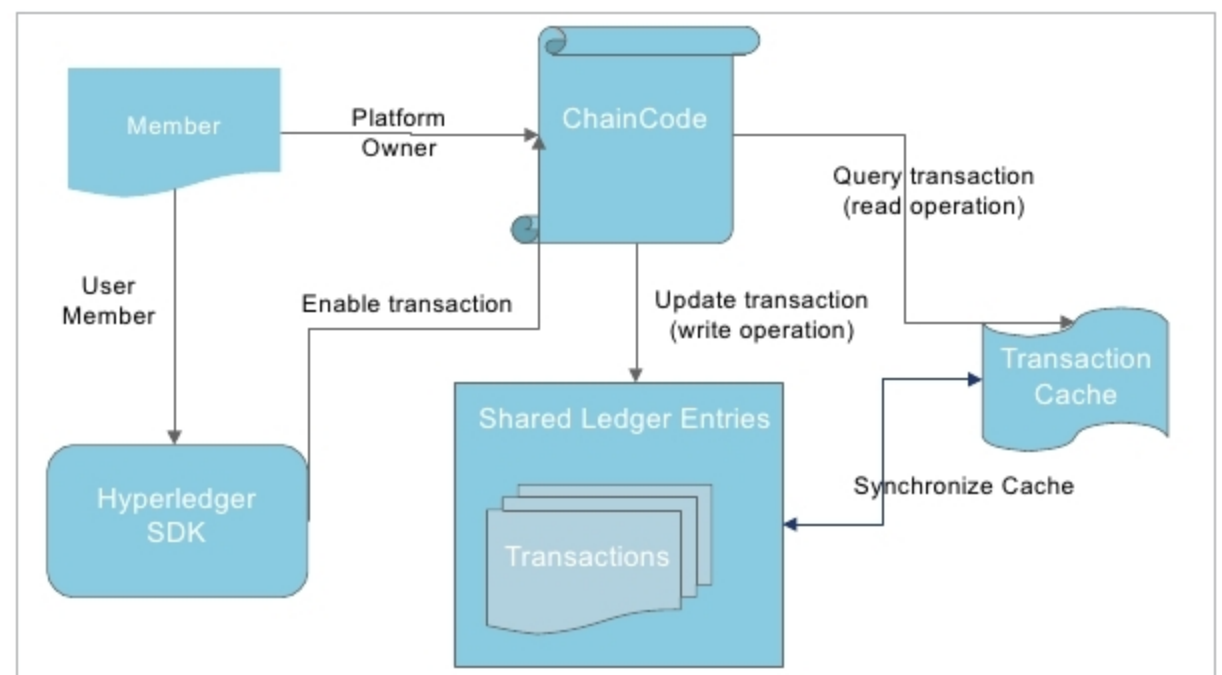


Figure 2: Chaincode implementation architecture

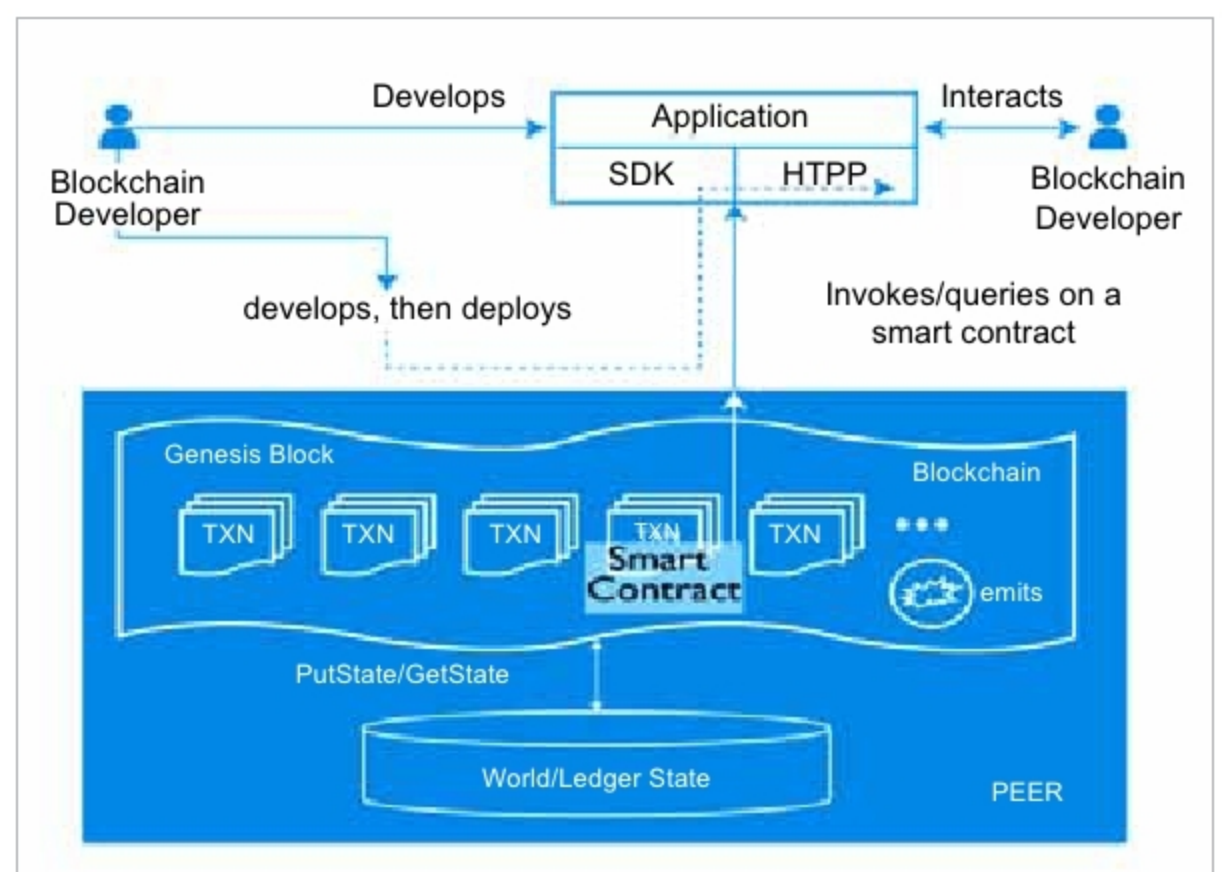


Figure 3: Hyperledger Fabric architecture